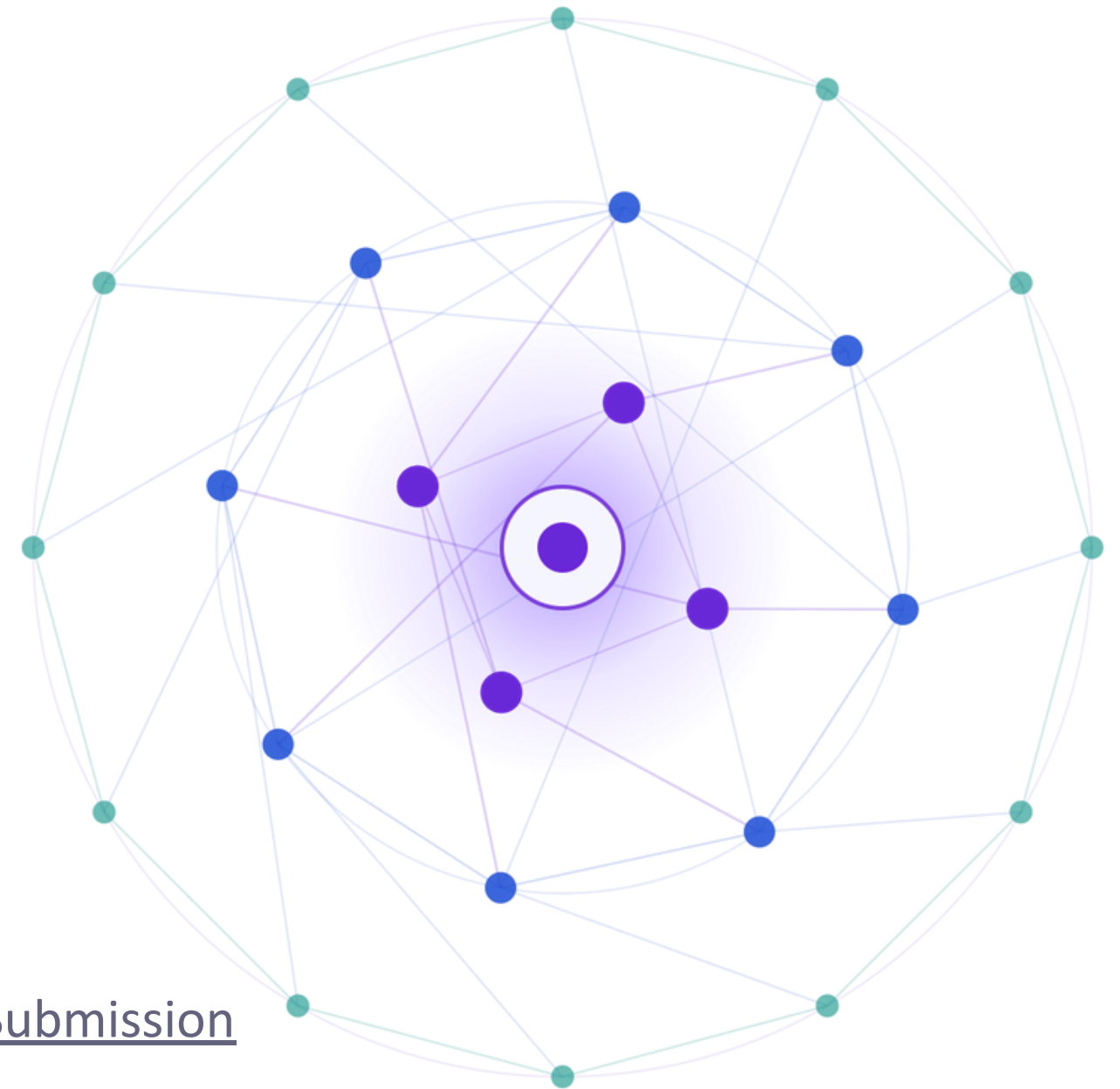


SAMSUNG PRISM

Generative AI Hackathon

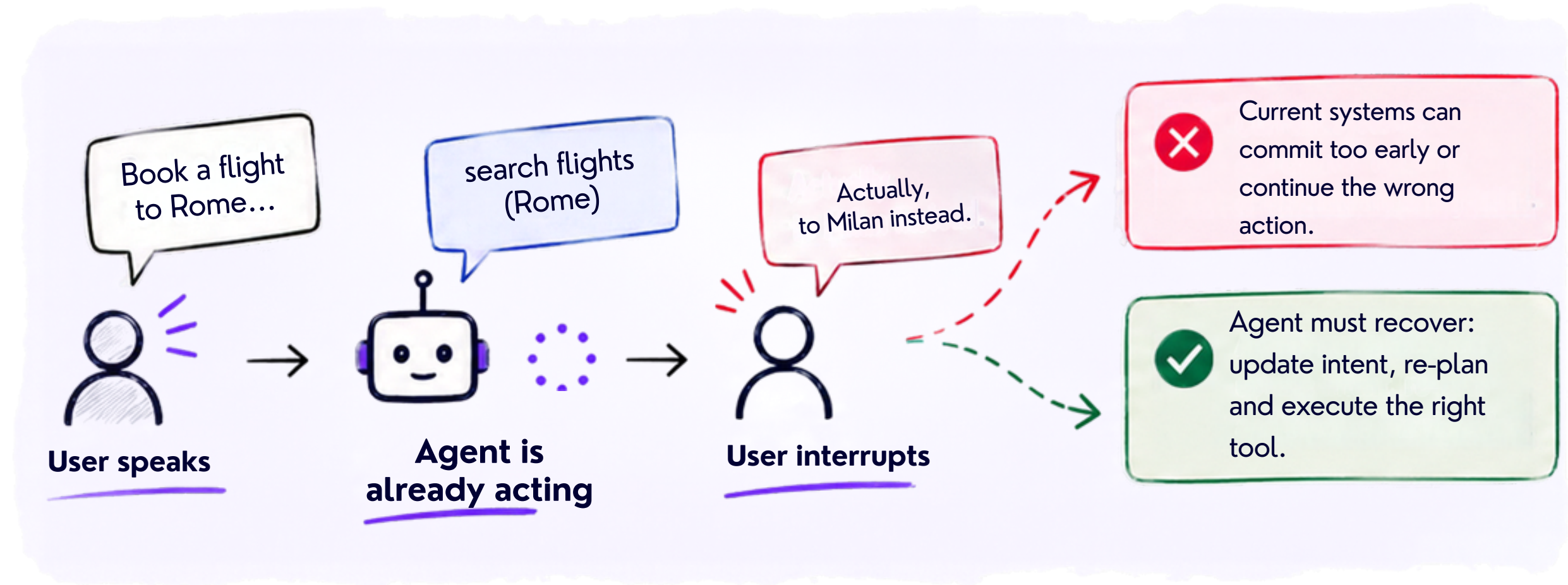
3rd Edition 2026 – 27

- Theme ID -05
- Team Name -**Genesis**
- College Name -**Thapar Institute of Engineering and Technology**
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- Member Name & Email 3- **Rishabh Dutt, rdutt_be24@thapar.edu**
- Member Name & Email 4- **Sukhesh Garg, sgarg3_be24@thapar.edu**
- Submission Github link - <https://github.com/Rishabh200729/SamsungPrismSubmission>



Interruptible Real-Time Voice Agents

A voice agent that can be interrupted, understands changes in real time, and takes the right action even while it is already speaking, reasoning or using tools.



Interrupt & update

Handle user changes mid-utterance while still processing.



State consistency

Keep tool arguments and context aligned with the latest intent.



Coordinated tool execution

Manage ongoing and new actions in real time.



Low end-to-end latency

Respond and complete tasks quickly despite interruptions.



Judged on Full-Duplex-Bench v3 (60%)

Tool-selection F1 | Argument accuracy | Strict pass rate | Latency



Evaluates real human speech with interruptions, self-corrections and multi-step tool calls.

Lin et al., 2026

Existing Solutions & Gaps

FDB-v3 exposes where current real-time voice systems still struggle when users interrupt, change intent and require multi-step actions.

Existing System	What happens (from FDB-v3)	Gap exposed	What must be solved
 GPT-Realtime	Best overall strict pass rate is only 0.600 (hard subset 0.433).	 Self-corrections and intent changes do not reliably propagate into action.	 State-consistent tool arguments and reliable re-planning.
 Gemini Live 3.1	In travel_19, tool call happened 2.27 s before the user finished and locked in "Rome".	 Can commit to tools too early before the user completes their intent.	 Better interruption detection and intent update before execution.
 Ultravox	47.9% of turns talk over the user; 88.0% filler.	 Speech overlap can become conversational interference.	 True interruption handling and turn control.
 Cascaded	Task completion latency is 10.12 s .	 End-to-end completion remains slow for multi-step tasks.	 Low-latency asynchronous execution and faster recovery.

Key Takeaways from Existing Solutions

1

Interruption support exists, **but correctness after** interruption is still limited.

2

Systems often commit to tools too early or fail to update intent consistently.

3

Speech overlap and turn management remain challenging in real conversations.

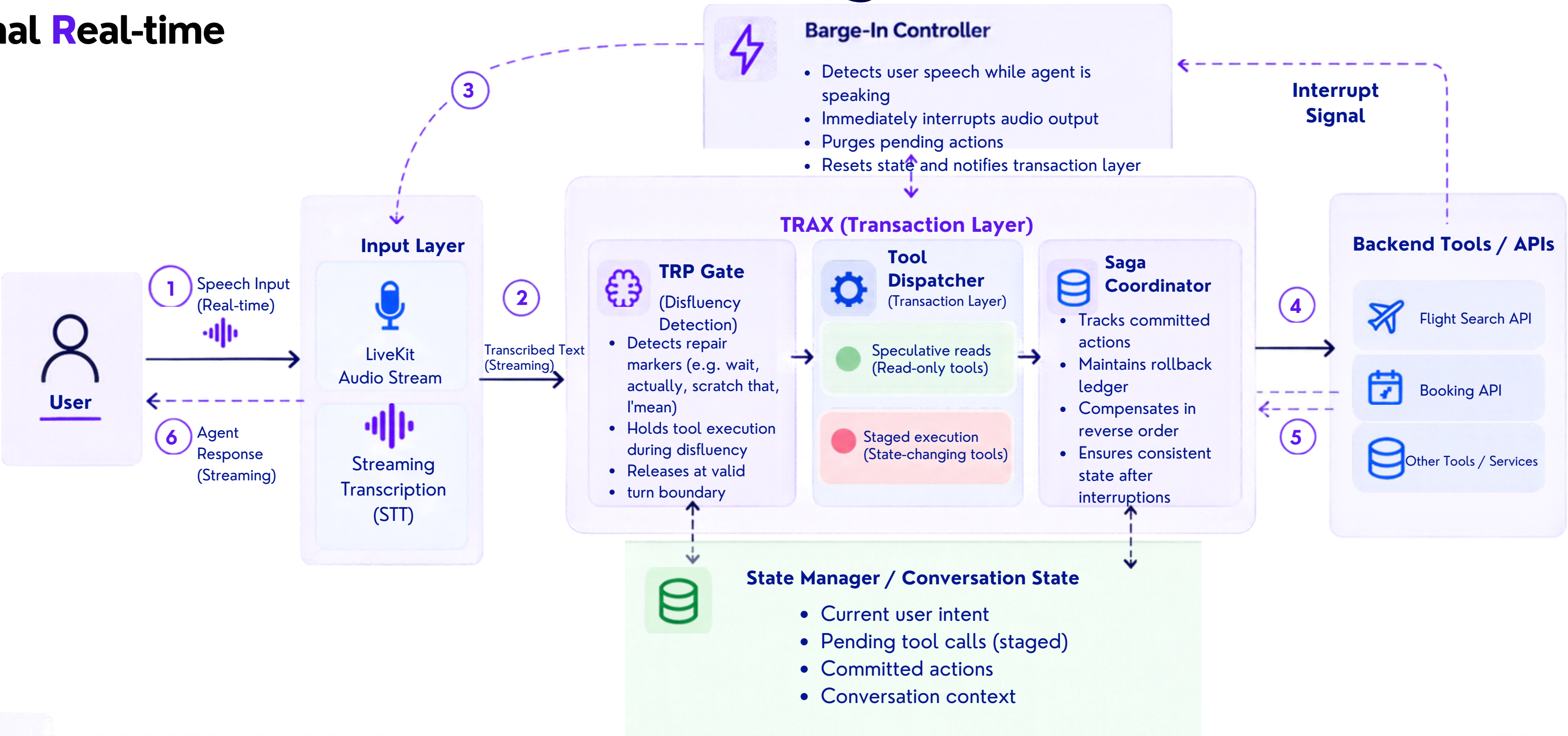
4

End-to-end task completion is still slow for multi-step actions.

Our Solutions & Architecture Diagram

TRAX: Transactional Real-time Agent Execution

TRAX is a voice-native transaction layer for interruptible real-time agents that understands when to act, when to wait, and how to safely recover after interruptions.



Key Components



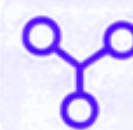
Disfluency-aware TRP Gate

Monitors streaming speech for repair markers and delays tool execution until a valid turn boundary.



Transactional Tool Execution

Executes read-only calls speculatively, stages state-changing actions and discards stale intent before commit.



Zero-lockout Intervention Handling

User speech detected while the agent is speaking immediately interrupts output, purges pending actions and resets the state.



Saga Rollback for Safe Recovery

Committed mutations are compensated in reverse order to prevent duplicate or harmful actions.

TRAX In-Car Navigation - Demo Walkthrough

[0.11s] =SYS= === Scenario 1: Driver requests a route ===

[0.11s] DRIVER "Navigate to the downtown hotel."

[0.11s] GATE -> TRP_CONFIRMED

[0.11s] MODEL emits update_destination(destination='downtown hotel')

[0.30s] AGENT "Routing you to downtown hotel via Main Street. ETA is 5 minutes with clear roads. Stay on Main Street for 1440 m, then continue straight."

TRAX IN-CAR INTELLIGENT NAVIGATION SYSTEM

ACTIVE DESTINATION: downtown hotel (3.6 km | 5 min)

ROUTE GUIDANCE: via Main Street (clear roads)

TRAX TRP GATE: TRP_CONFIRMED

TRANSACTION LEDGER:

1. update_destinationNone-> downtown hotelCOMMITTED

[0.81s] =SYS= === Scenario 2: Mid-sentence self-correction (last-writer-wins) ===

[0.81s] DRIVER "Take me to the city mall..."

[1.02s] MODEL emits update_destination(destination='city mall')

[1.33s] DRIVER "...no wait, actually the airport."

[1.33s] GATE -> REPAIRING

[1.59s] MODEL emits update_destination(destination='airport')

[2.30s] GATE -> TRP_CONFIRMED

[2.30s] VEHICLE active destination = 'airport' (stale call: superseded)

[2.30s] =SYS= === Scenario 3: Live barge-in -> saga rollback (same agent turn) ===

[2.30s] AGENT ""Sure, I caught that - routing you to the airport instead. Via Airport Expressway, ETA 28 minutes with light traffic. Airport Expressway for 7360 m, then continue straight.""

[2.55s] DRIVER "Hold on - take me to central station instead." [BARGING IN]

[2.55s] GATE -> ABORTED

[2.56s] VEHICLE active destination = 'downtown hotel' (airport reroute rolled back)

[2.56s] GATE -> REPAIRING

[2.86s] MODEL emits update_destination(destination='central station')

[3.53s] GATE -> TRP_CONFIRMED

[3.53s] AGENT ""Got it - routing to Central Station via Central Boulevard. ETA 9 minutes, light traffic. Stay on Central Boulevard for 1440 m, then continue straight.""

DEMO AND PRODUCT WALKTHROUGH

...mission — -zsh ...

...mission — -zsh ...

...mission — -zsh ...

...mission — -zsh ...

...mission — -zsh ...

...ent.run cons...

to find the best value. You might also want to look into factors like layovers, flight duration, and potential airline per

Fortable journey."}

3:12.481 DEBUG trax.trp_gate trp_gate: silence 300ms ≥ threshold but buffer tail incomplete, holding

3:12.689 DEBUG trax.trp_gate trp_gate: silence 500ms ≥ threshold but buffer tail incomplete, holding

3:13.095 DEBUG trax.trp_gate trp_gate: silence 900ms ≥ threshold but buffer tail incomplete, holding

3:40.954 INFO prism.agent DEBUG: User query ended at 1790785120.954937

3:40.959 INFO trax.barge_in barge_in: user speech onset (agent_speaking=False, handling=False)

3:45.720 DEBUG livekit.agents conversation_item_added {"role": "user", "lk.pii.text": "एकजुली अस्मि फ्र इंड मी द फ्लइट्स टू मुंबई..."} (tier=0,

3:45.724 INFO trax.trp_gate trp_gate: final token + complete buffer 'ी अस् सो फ्र इंड मी द फ्ल इ ट्स टू मुंबई...' (tier=0,

3:45.729 DEBUG trax.trp_gate quiescence=n/a) → TRP_CONFIRMED

3:45.732 INFO trax.too...ispatcher trp_gate: LISTENING → TRP_CONFIRMED

3:47.717 INFO prism.agent dispatcher: TRP_CONFIRMED - committing 0 speculative + 0 staged

3:47.724 INFO trax.barge_in DEBUG: User query ended at 1790785127.717724

3:47.727 DEBUG trax.trp_gate barge_in: user speech onset (agent_speaking=False, handling=False)

3:47.953 INFO trax.trp_gate trp_gate: reset_for_new_turn (was TRP_CONFIRMED)

3:47.956 DEBUG trax.trp_gate trp_gate: CORRECTION (tier-1) detected in 'No wait' → REPAIRING (VAD inflated to 900ms)

3:47.959 INFO trax.too...ispatcher trp_gate: LISTENING → REPAIRING

3:48.765 INFO trax.trp_gate dispatcher: state=REPAIRING - all commits suspended

3:48.770 DEBUG trax.trp_gate trp_gate: tier-1 quiescence window (811ms) elapsed → TRP_CONFIRMED

3:48.774 INFO trax.too...ispatcher trp_gate: REPAIRING → TRP_CONFIRMED

3:51.191 DEBUG livekit.agents dispatcher: TRP_CONFIRMED - committing 0 speculative + 0 staged

3:51.199 DEBUG livekit.agents conversation_item_added {"role": "user", "lk.pii.text": "No wait actually find the flight to Bangalore."}

3:51.208 INFO trax.too...ispatcher executing tool {"function": "search_flights", "lk.pii.arguments": "{\\"destination\\": \\"Bangalore\\", \\"date\\": "2026\\"}"}, "speech_id": "speech_c307d55624bd"} (tier=0,

3:51.212 DEBUG trax.too...ispatcher dispatcher: intercepted search_flights({"destination": "Bangalore", "date": "October 5, 2026"}) [READ_ONLY] - current state=TRP_CONFIRMED

3:51.217 DEBUG trax.too...ispatcher dispatcher: search_flights speculative task started

3:51.222 INFO trax.too...ispatcher dispatcher: TRP already confirmed, committing search_flights immediately

3:51.227 DEBUG livekit.agents dispatcher: COMMITTED (read-only) search_flights → {"status": "success", "flights": [{"flight_id": "FL123", "pr...

3:52.688 INFO trax.barge_in tools execution completed {"speech_id": "speech_c307d55624bd"}

3:52.693 DEBUG trax.trp_gate barge_in: user speech onset (agent_speaking=False, handling=False)

3:53.761 INFO trax.barge_in trp_gate: reset_for_new_turn (was TRP_CONFIRMED)

trax.barge_in barge_in: agent speech started

ok Air Microphone _ _ _ _ _ _ _ _ _ _

Tools and tech stack used

The runtime is split between realtime media, intelligence, transactional control, and benchmark evaluation.

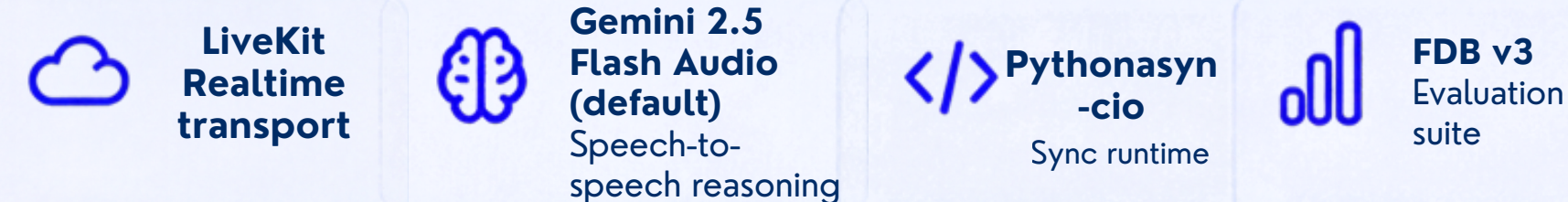


What is custom vs runtime foundation?

☆ CUSTOM TRAX CONTROL



🗄️ EXTERNAL /RUNTIME FOUNDATION



Real-World Use Cases

TRAX enables safe, responsive and interruption-ready voice agents across high-value domains.



Same TRAX architecture
Multiple real-world applications

01

Automotive & In-Car

High-speed, safety-critical navigation

Take me to the
Central City Mall...

No, actually the
airport instead.



Handles route
changes mid-
utterance



Prevents wrong
route execution



Keeps driver
focused and
safe

02

Travel & Bookings

Stateful multi-step transactions



Flight to Delhi

Fri, Oct 6

Sat, Oct 7



Book a flight
to Delhi...

Actually,
Saturday instead.



Updates intent
and slots in real
time



Avoids duplicate
bookings and
charges



Completes
correction action
confidently

03

Enterprise Support

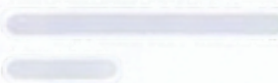
Complex, stateful workflows



Update the address
for my account...

No, use my office
address instead.

Account Update



Office Address



Handles multi-step
tool sequences with
corrections



Prevents stale
or incorrect
updates



Improves
customer
experience



Broader Impact

TRAX brings reliable, real-time, interruption-ready voice intelligence to real-world, high-stakes scenarios.



Safer
interactions



Higher
productivity



Better
user experiences

INNOVATION, HIGHLIGHTS, RESULTS AND LIMITATIONS



CORE INNOVATION

TRAX unifies identity graph, AI decisions, and real-time orchestration to personalize journeys and automate actions across channels.

360°

AI-NATIVE

predictive decisions
in real time

OMNICHANNEL

journeys that adapt
everywhere



TECHNICAL HIGHLIGHTS & DIFFERENTIATORS

Microservices architecture built for scale and resilience
Real-time event streaming with sub-second latency
Privacy-by-design with consent and governance
Open APIs and composable ecosystem

SUB-SECOND

decision latency

99.9%+

platform availability

API-FIRST

open and extensible
by design



BUSINESS & ENTERPRISE IMPACT

Stronger customer engagement and loyalty
Higher conversion and revenue uplift
Operational efficiency and cost optimization
Faster time to-market for new initiatives

18%+

increase in
conversion

25%+

improvement in
marketing ROI

30%+

reduction in
manual effort



VERIFICATION & RESULTS

Proven outcomes across pilot and production deployments
Validated by A/B testing and holdout analysis
Measurable impact on key business KPIs
Continuous monitoring and model improvement

2.5X

lift in customer
engagement

15%+

revenue uplift
per user

HIGH

statistical
confidence



ENGINEERING LIMITATIONS & BOUNDARIES

Data quality and identity
resolution impact accuracy

Dependent on third-
party integrations and
latency

Model performance
varies by market and
segment

Not a real-time
system of record

Subject to privacy,
consent, and regulatory
constraints

WHAT'S NEXT

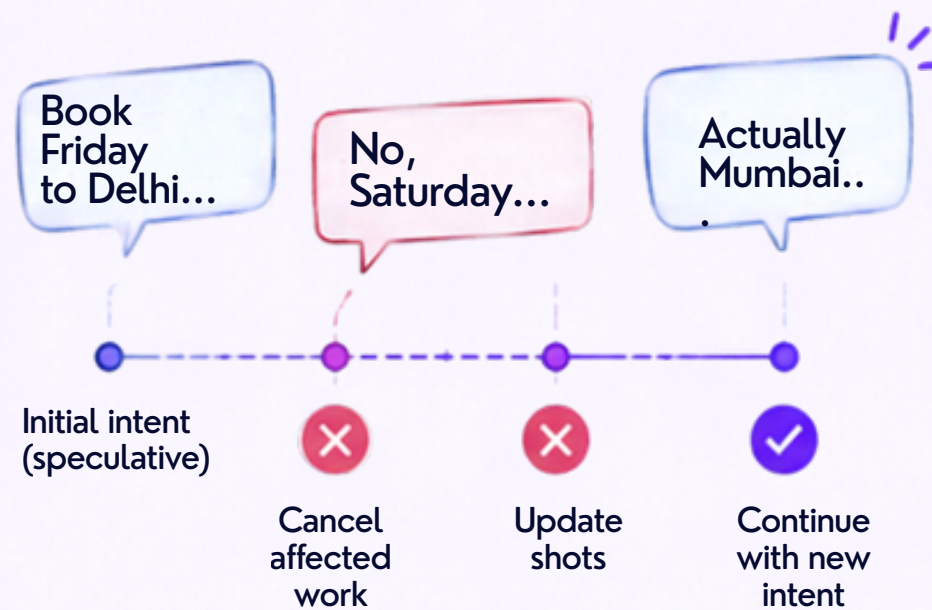
From interruptible → anticipatory → domain-general

01



Deeper self-correction

Track multiple semantic repairs and selectively invalidate stale work.

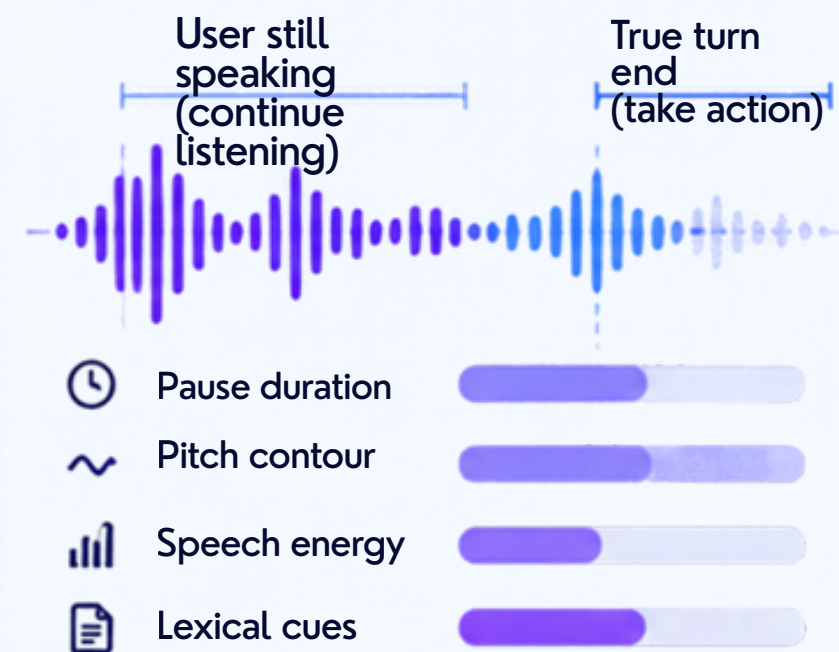


02



Prosody-aware turn prediction

Combine speech timing, prosody and lexical cues to distinguish hesitation from true turn completion.



03



Confidence-aware speculation

Start safe read-only work before end-of-turn; commit only after intent stabilizes.



04



Domain-general transactional execution

Extend compensation, idempotency and rollback beyond navigation to finance, housing and other stateful voice workflows.



Reactive interruption handling



TRAX

Anticipatory execution



General transactional voice intelligence